AUTOMATED LEFT ATRIAL SEGMENTATION FROM MAGNETIC RESONANCE IMAGE SEQUENCES USING DEEP CONVOLUTIONAL NEURAL NETWORK WITH AUTOENCODER

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ABSTRACT

This study presents a novel automated algorithm to segment the left atrium (LA) from 2, 3 and 4-chamber long-axis cardiac cine magnetic resonance image (MRI) sequences using deep convolutional neural network (CNN). The objective of the segmentation process is to delineate the boundary between myocardium and endocardium and exclude the mitral valve so that the results could be used for generating clinical measurements such as strain and strain rate. As such, the segmentation needs to be performed using open contours, a more challenging problem than the pixel-wise semantic segmentation performed by existing machine learning-based approaches such as U-net. The proposed neural net is built based on pre-trained CNN Inception V4 architecture, and it predicts a compressed vector by applying a multi-layer autoencoder, which is then back-projected into the segmentation contour of the LA to perform the delineation using open contours. Quantitative evaluations were performed to compare the performances of the proposed method and the current state-of-the-art U-net method. Both methods were trained using 6195 images acquired from 80 patients and evaluated using 1515 images acquired from 20 patients where the datasets were obtained retrospectively, and ground truth manual segmentation was provided by an expert radiologist. The proposed method yielded an average Dice score of 93.1% and Hausdorff distance of 4.2 mm, whereas the U-net yielded 91.6% and 11.9 mm for Dice score and Hausdorff distance metrics, respectively. The quantitative evaluations demonstrated that the proposed method performed significantly better than U-net in terms of Hausdorff distance in addition to providing open contour-based segmentation for the LA.

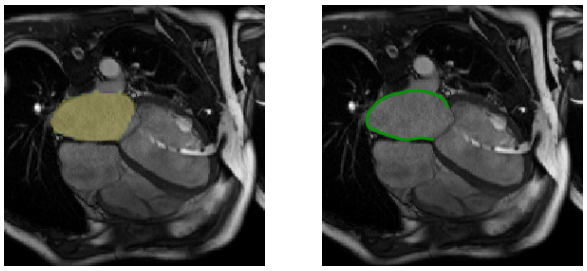
Index Terms— Magnetic resonance imaging (MRI), left atrial segmentation, autoencoders, convolutional neural networks (CNN).

I. INTRODUCTION

The assessment of the left atrial (LA) function is important in the clinical management of cardiac disease. Several clinical studies emphasize the diagnostic and prognostic values of the LA functional assessment[1], [2], [3]. While both the computed tomography (CT) and magnetic resonance imaging (MRI) can be used for the anatomical assessment of the LA, the functional assessment of the LA is often performed using MRI. Such functional assessment requires segmenting the LA from a large number of images in cardiac MRI, which is a tedious and complex task for clinicians who have to manually delineate the chamber. As such, automated delineation of the LA is of great interest.

Automated and semi-automated cardiac chamber delineation methods have been reviewed in [4], and most existing methods focus on the segmentation of the left and right ventricles. However, the automated segmentation of the LA is more challenging than left or right ventricles due to the following reasons: 1) the size of the LA is relatively small compared to the left or right ventricles; 2) boundaries are not clearly defined when the blood pool of the LA goes into the pulmonary veins; and 3) the shape variability of the LA is significant between different subjects [5]. Only a few studies that address the problem of segmenting the LA from MRI [5], [6], [7], [8]. However, these methods either relied on static 3D MRI scans [5], [6], [7] which are not suitable for functional analysis or included the mitral valve in the segmentation [8]. To compute clinical measurement such as strain and strain rate related to the LA function, the segmentation approach should delineate the boundary between the endocardium and myocardium and exclude the mitral valve. As such, the segmentation should be performed using open-contours which is a more challenging problem than the popular pixel-wise semantic segmentation performed using existing machine learning-based algorithms [9], [8]. Refer to Fig. 1 for an example showing the difference between segmentation using open-contours and semantic segmentation.

This study proposes a novel open-contour based segmentation approach to delineate the LA from MRI sequences using a deep convolutional neural network (CNN) architecture. The proposed approach incorporates autoencoders to detect the segmentation contours directly from the raw MR images. Previous unsupervised approaches such as principal component analysis based linear reconstruction [10] are not suitable since they are unable to preserve the highly nonlinear relationships when projected to low-dimensional space. The proposed automated segmentation using open-



(a) Semantic segmentation (b) Open contour segmentation

Fig. 1: Example showing the difference between pixel-wise semantic segmentation (a) and delineation using an open contour (b) of the LA. The results of the semantic segmentation includes the mitral valve. The computation of the strain and strain-rate measurements of the left atrial function require exclusion of mitral valve which could only be performed using open contours.

contours exclude the mitral valve and could be used for computing strain and strain rate measurements.

II. METHODOLOGY

II-A. Proposed Deep ConvNet Architecture

This study utilizes the Inception-V4 [11] pre-trained on ImageNet Large Scale Visual Recognition Competition (ILSVRC) datasets and updates the network parameters to be used with MRI datasets. The first convolutional layer and the last fully connected layers are modified by changing the initial weights according to our problem. The Inception-V4 was trained on RGB color images with 3 channels which are different from the MRI dataset used in this study consisting of grey-scale values. Therefore, 3 channels of the CNN were reduced to 1 channel. The ImageNet version of the algorithms comes with a softmax layer as the last layer with 1000 categories. However, it does not include an approach to predict the object boundaries and, therefore, the task of predicting segmentation contour (*i.e.,* 482×1 vector) should be developed for the proposed problem. As such, the last layer (softmax layer) of the pre-trained network is truncated and replaced with a new softmax layer with the dimension of 482. In the proposed neural net approach, the mean squared error (MSE) is used as the loss function and Adam version of the stochastic gradient descent is used as the optimizer [12].

II-B. Dimensionality Reduction by Autoencoders

The autoencoders and its deep version as traditional dimensionality reduction methods have achieved great success via the powerful representability of neural networks. However, they only use each instance to reconstruct itself and ignore to explicitly model the data relation to discover the underlying effective manifold or the feature space in the dataset [13]. An autoencoder generally consists of two parts, namely, an encoder which transforms the input to a hidden code and a decoder which reconstructs the input from hidden code.

The size of hidden layer is smaller than the input layer in autoencoders and is 50 in our case. By reducing the hidden layer size, we force the network to learn the important features of the dataset. The main reason for using autoencoders in our problem is to reduce the dimensionality and reconstruct the original contour from the hidden representation. The non-linearity property of the autoencoders captures the spatial information regarding the shape and sharp edges of the segmentation contour. Before computing the hidden representation, we impose the non-linear activation function, rectified linear unit (ReLU), to introduce non-linearities in the linear output.

In our implementation, at first we used a multi-layer autoencoder and trained it with the segmentation contour, *i.e.* the 482×1 vector (241 x-coordinates and 241 y-coordinates) and the bottleneck dimension is 50×1. Then the CNN is trained with the target as the bottleneck vector of the autoencoder (i.e. 50×1 latent representation). After CNN's prediction, we attach the decoder part of the autoencoder and CNN's prediction (*i.e.,* the 50×1 vector) is passed through the decoder to reconstruct the original 482×1 vector which produces the segmentation contour. We train both the models end-to-end as shown in Fig. 2. In the autoencoders, we impose the non-linear activation function ReLU to capture complicated properties such as shape and size related to the underlying segmented structure.

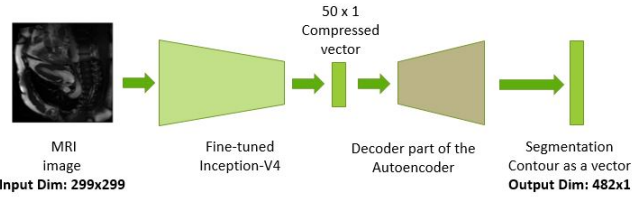


Fig. 2: Components of the proposed neural net approach which consists of a fine-tuned CNN architecture combining the decoder part of the auto-encoder.

III. EXPERIMENTAL SETUP AND RESULTS

III-A. Datasets

The proposed method and U-net approach were trained and tested over 100 MRI patient datasets acquired at the University Alberta Hospital. The study was approved by the Health Research Ethics Board of the University of Alberta. The details of dataset can be found in [8], [14]. The evaluations were performed on 1515 images acquired from 20 patients in comparison to the manual segmentation of the LA from the 2, 3 and 4-chamber long-axis MR image sequences. The ground truth manual segmentation

was initially performed by a medical student using a semi-automated software and the contours were corrected by an experienced radiologist.

III-B. Data Preprocessing

The MR images are of different dimensions or sizes. Therefore, the images have to be resized to fit the input shape of the network. The input to the proposed network is a 2D MR image of size 299×299 and the output is the segmentation contour, *i.e.,* the stacked (x, y) coordinate points of the contour. We used 241 pairs of (x, y) points and stacked them to form a vector of size 482×1. In the proposed approach, the CNN directly predicts the latent representation of the segmentation using an open contour. Therefore, resizing the raw images in any dimension does not affect the prediction of the contour positions. Each input image and the segmentation mask for U-net is resized to 256×256.

III-C. Implementation Details

We input a 299×299×1 of grayscale MRI image into the InceptionV4 to predict a 50×1 response vector using autoencoder. The training was performed using 250 epochs and a batch size of 6. The last layer is a dense layer that uses the MSE between the actual and the predicted value as the loss function and illustrated in (1). Adam version of the stochastic gradient descent [12] is used as the optimizer. All the experiments with the neural network were implemented using Keras API [15] with TensorFlow and on two NVIDIA GeForce GTX 1080 GPU processors. It took around 6 hours to train the network. The prediction took 5 ms per image.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y_i})^2 \tag{1}$$

III-D. Quantitative Evaluation Metrics

For evaluation, we used Dice Coefficient (DC) and the Hausdorff Distance (HD) between the ground truth segmentation and the segmentation obtained using the automated methods. A higher DC value is associated with a better segmentation result. The lower value of Hausdorff distance corresponds to better segmentation. In this study, the HD was measured by comparing the auto contours with the open contours representing the LA walls drawn manually whereas the DC was computed by comparing the regions enclosed by the auto and manual contours.

III-E. Reliability of an Algorithm

We also computed the *reliability* of the above algorithms by evaluating the *reliability function* R defined below. The reliability function is the complementary cumulative distribution of the obtained DC and is defined for each $d \in [0, 1]$ as the probability of obtaining the DC greater than d over the entire dataset [16]. $R(d) = Pr(\text{dice score} > d) =$ (number of images segmented with DC greater than d)/(total number of images). This measures how reliable an algorithm in at least achieving accuracy d.

Table I: Evaluation of automated segmentation results by the proposed method and U-net in terms of Dice Coefficient (DC) and Hausdorff Distance (HD) in comparison to expert manual segmentation over 1515 images. The Mean ± Standard Deviation values are reported for each metric. The higher the Dice coefficient or the lower the Hausdorff distance the better the results.

	Dice Coefficient (%)		Hausdorff Distance (mm)	
View	Our Method	U-net [14]	Our Method	U-net [14]
2-ch	94.1 ± 3.3	93.4 ± 3.0	4.0 ± 3.2	7.7 ± 6.4
3-ch	92.2 ± 4.1	92.4 ± 4.0	4.7 ± 3.8	16.3 ± 22.9
4-ch	92.8 ± 4.0	88.9 ± 11.3	3.8 ± 3.1	11.8 ± 10.9

III-F. Experimental Results

Table I reports the DC and the HD between manual segmentation and the segmentation by automated methods. The results are computed over 1515 images of the LA. The proposed CNN with autoencoder performed significantly better than the current state-of-the-art method U-net in terms of Hausdorff distance for 2, 3 and 4 chamber sequences. The performance of the proposed algorithm is better than U-net in terms of Dice score for 2 and 4 chamber sequences. In Fig. 4, the highest accuracy (98%), 50th percentile (90%) and the lowest accuracy (80%) of Dice scores achieved by the proposed method are shown visually, where the green, yellow and red contours depict the segmentation results by the proposed method, U-net approach and ground truth, respectively.

The reliability curves for all the methods are shown in Fig. 3. The figure demonstrates that the proposed method yielded delineations with a higher reliability than the U-net.

IV. CONCLUSION

This study proposes a novel method for the automatic segmentation of the left atrium (LA) using deep CNN augmented with autoencoder. The LA segmentation is very challenging even for experts as there is no strong line of demarcation for segmenting the structure in MRI. By applying autoencoder, we imposed a strong structured regression approach to the proposed model. The proposed approach incorporates autoencoders to obtain delineation using the open contours. The open contour based segmentation is essential in the computation of clinical measurements such as strain and strain-rate that exclude mitral valve in the computation, whereas, existing machine learning-based methods such as U-net rely on semantic segmentation which may include the mitral valve and not suitable for generating such clinical measurements.

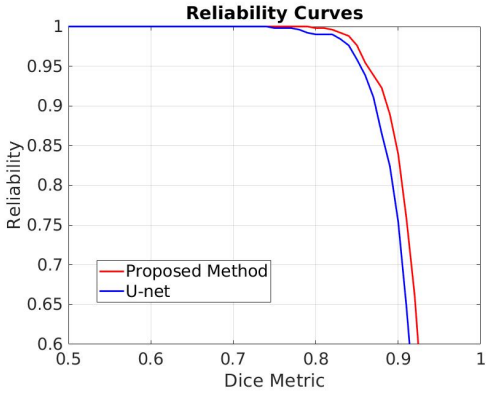
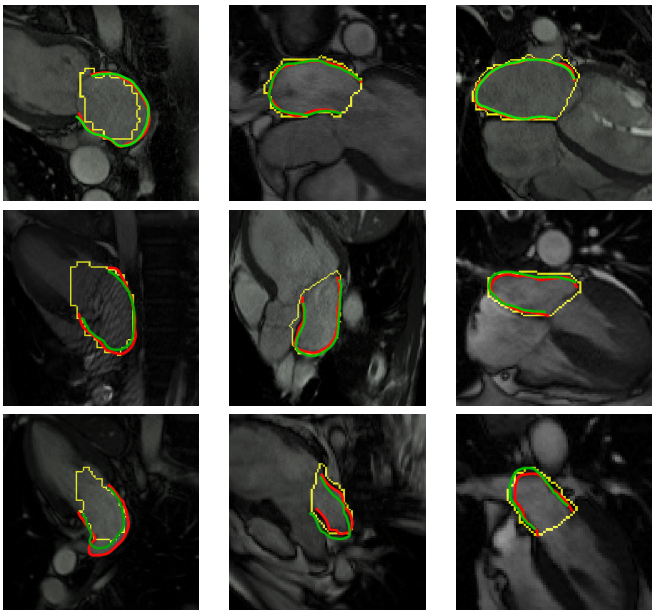


Fig. 3: Reliability versus Dice metric for the proposed method and U-net evaluated over 1515 MRI images. In addition to providing open contour based segmentation, the proposed method yielded more reliable segmentation results than the U-net approach.



(a) 2-chamber (b) 3-chamber (c) 4-chamber

Fig. 4: The automated segmentation results with the highest (1st row), 50th percentile (2nd row), lowest (3rd row) Dice score values for (a) 2-chamber (1st column), (b) 3-chamber (2nd column), (c) 4-chamber (3rd column) MRI views. The red, green and yellow contours represent the ground truth, automated segmentation by the proposed method and the boundary corresponds to the semantic segmentation by U-net, respectively.

The proposed approach yielded an average Dice score of 93.1% and an average Hausdorff distance of 4.2 mm in comparison to the ground truth segmentation. This study demonstrates that the proposed method achieves high conformance with manual segmentation and is a promising tool for the fully automated delineation of the LA.

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